



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

This project analyzes SpaceX Falcon 9 launch data to understand the factors influencing first-stage landing success. Using data collected from the SpaceX REST API and web-scraped sources, the dataset was cleaned, wrangled, and enriched to support exploratory data analysis, interactive analytics, and predictive modeling.

Key components of the project include:

- Collecting launch data from the SpaceX API and Wikipedia tables
- Wrangling and cleaning data, including handling missing values and filtering Falcon 1 launches
- Performing EDA using visualization and SQL
- Building interactive analytics using Folium and Plotly Dash
- Developing classification models to predict landing success

The final outcome is a set of insights, dashboards, and predictive models that support decision-making for **Space Y**, aiming to estimate launch costs and understand SpaceX's reusability patterns.

Introduction

Project Background and Context

The commercial space industry is rapidly expanding, with companies like Virgin Galactic, Rocket Lab, Blue Origin, and SpaceX driving innovation. SpaceX stands out due to its ability to reuse the Falcon 9 first stage, significantly reducing launch costs.

Understanding whether the first stage will land successfully is crucial because:

- A successful landing reduces launch cost
- Landing success depends on multiple factors (payload, orbit, launch site, etc.)
- Predicting landing success helps estimate launch pricing

Introduction

Problems We Want to Find Answers To

- What factors influence Falcon 9 first-stage landing success?
- How do launch site, payload mass, orbit type, and flight number correlate with outcomes?
- Can we build a predictive model to determine landing success?
- What insights can interactive analytics reveal about launch patterns?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Injection via SpaceX-API and web scraping
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Data was collected from two sources:
 - SpaceX REST API: using the endpoint `api.spacexdata.com/v4/launches/past`
 - Web Scraping: using the Python BeautifulSoup package to web scrape some HTML tables

Data Collection – SpaceX API

Process summary:

- Performed GET requests to retrieve past launch data
- Parsed JSON responses
- Normalized nested JSON into tabular format using `json_normalize`
- Retrieved additional details (booster, payload, launchpad, core) using secondary API calls
- Filtered out Falcon 1 launches
- Handled missing values (e.g., replacing null PayloadMass with mean)

- <https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

[32]:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	Reu:
0	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	
1	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	
2	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	
3	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	
4	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
85	86	2020-09-03	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	2	True	True	True	5e9e3032383ecb6bb234e7ca	5.0	
86	87	2020-10-06	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	3	True	True	True	5e9e3032383ecb6bb234e7ca	5.0	
87	88	2020-10-18	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	6	True	True	True	5e9e3032383ecb6bb234e7ca	5.0	
88	89	2020-10-24	Falcon 9	15600.0	VLEO	CCSFS SLC 40	True ASDS	3	True	True	True	5e9e3032383ecb6bb234e7cc	5.0	
89	90	2020-11-05	Falcon 9	3681.0	MEO	CCSFS SLC 40	True ASDS	1	True	False	True	5e9e3032383ecb6bb234e7ca	5.0	

90 rows x 17 columns

Data Collection - Scraping

Process summary:

- Scraped Falcon 9 launch tables from Wikipedia
- Parsed HTML tables using BeautifulSoup
- Converted tables into Pandas DataFrames
- Cleaned and standardized scraped data
- <https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb>

2020 [edit]

In late 2019, [Gwynne Shotwell](#) stated that SpaceX hoped for as many as 24 launches for Starlink satellites in 2020,^[490] in addition to 14 or 15 non-Starlink launches. At 26 launches, 13 of which for Starlink satellites, Falcon 9 had its most prolific year, and Falcon rockets were second most prolific rocket family of 2020, only behind China's [Long March](#) rocket family.^[491]

[hide] Flight No.	Date and time (UTC)	Version, Booster ^[4]	Launch site	Payload ^[4]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21 ^[492]	F9 B5 Δ B1049.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success (drone ship)
Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations. ^[493]									
79	19 January 2020, 15:30 ^[494]	F9 B5 Δ B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test ^[495] (Dragon C205.1)	12,050 kg (26,570 lb)	Sub-orbital ^[496]	NASA (CTS) ^[497]	Success	No attempt
An atmospheric test of the Dragon 2 abort system after Max Q. The capsule fired its SuperDraco engines, reached an apogee of 40 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously slated to be accomplished with the Crew Dragon Demo-1 capsule ^[498] but that test article exploded during a ground test of SuperDraco engines on 20 April 2019. ^[419] The abort test used the capsule originally intended for the first crewed flight. ^[499] As expected, the booster was destroyed by aerodynamic forces after the capsule aborted. ^[500] First flight of a Falcon 9 with only one functional stage — the second stage had a mass simulator in place of its engine.									
80	29 January 2020, 14:07 ^[501]	F9 B5 Δ B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success (drone ship)
Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the fairing halves was caught, while the other was fished out of the ocean. ^[502]									
81	17 February 2020, 15:09 ^[503]	F9 B5 Δ B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Failure (drone ship)
Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km × 386 km (132 mi × 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage booster failed to land on the drone ship ^[504] due to incorrect wind data. ^[505] This was the first time a flight proven booster failed to land.									
82	7 March 2020, 04:50 ^[506]	F9 B5 Δ B1059.2	CCAFS, SLC-40	SpaceX CRS-20 (Dragon C112.3 Δ)	1,977 kg (4,359 lb) ^[507]	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
Last launch of phase 1 of the CRS contract. Carries <i>Bartolomeo</i> , an ESA platform for hosting external payloads onto ISS. ^[508] Originally scheduled to launch on 2 March 2020, the launch date was pushed back due to a second stage engine failure. SpaceX decided to swap out the second stage instead of replacing the faulty part. ^[509] It was SpaceX's 50th successful landing of a first stage booster, the third flight of the Dragon C112 and the last launch of the cargo Dragon spacecraft.									
83	18 March 2020, 12:16 ^[510]	F9 B5 Δ B1048.5	KSC, LC-39A	Starlink 5 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Failure (drone ship)
Fifth operational launch of Starlink satellites. It was the first time a first stage booster flew for a fifth time and the second time the fairings were reused (Starlink flight in May 2019). ^[511] Towards the end of the first stage burn, the booster suffered premature shut down of an engine, the first of a Merlin 1D variant and first since the CRS-1 mission in October 2012. However, the payload still reached the targeted orbit. ^[512] This was the second Starlink launch booster landing failure in a row, later revealed to be caused by residual cleaning fluid trapped inside a sensor. ^[513]									
84	22 April 2020, 19:30 ^[514]	F9 B5 Δ B1051.4	KSC, LC-39A	Starlink 6 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success (drone ship)

Data Wrangling

[13]:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004

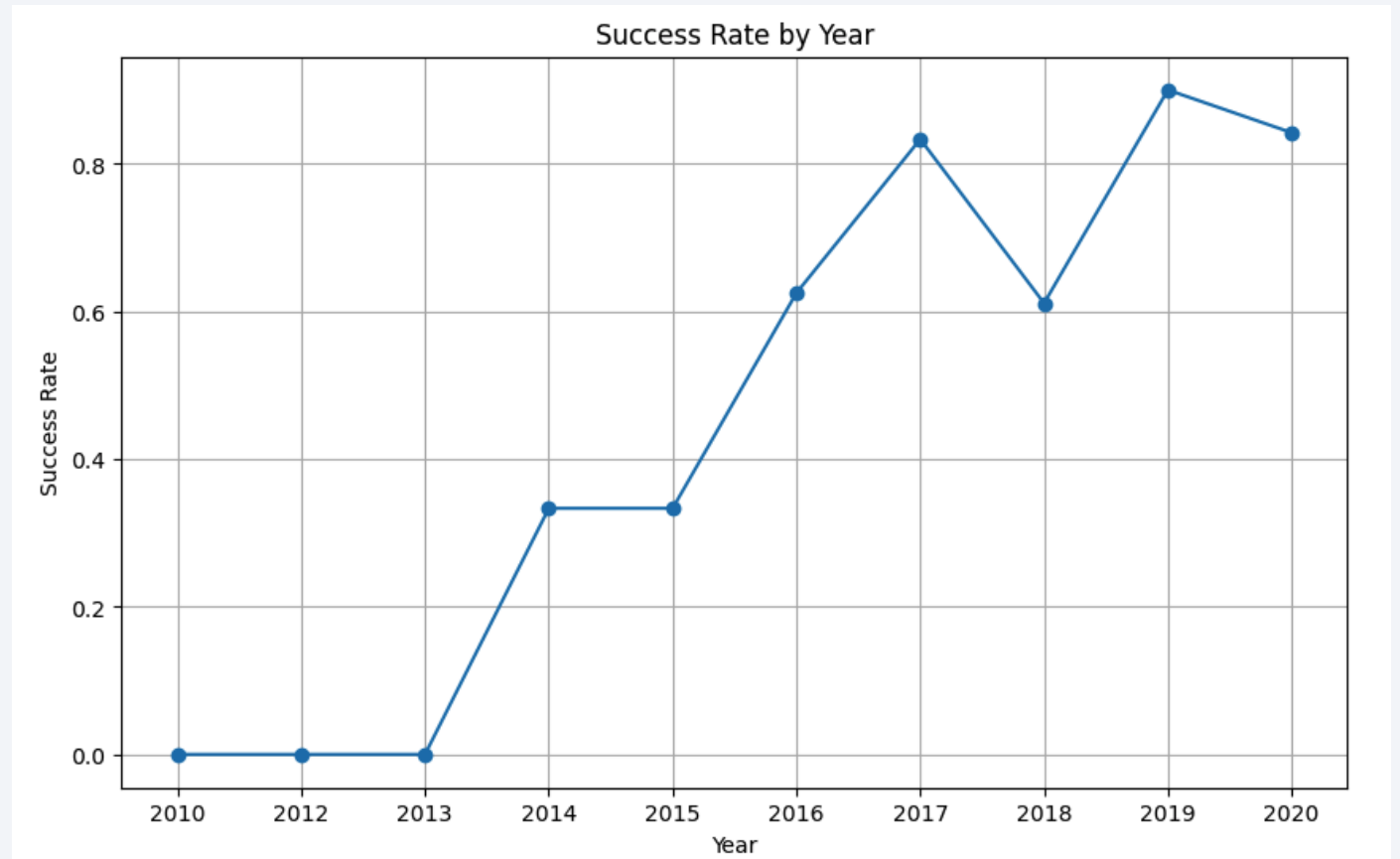
Key steps:

- Reviewed attributes such as Flight Number, Date, Booster Version, Payload Mass, Orbit, Launch Site, Outcome, Grid Fins, Legs, Landing Pad, Block, Serial, Coordinates
- Converted landing outcomes into binary classes (0 = failure, 1 = success)
- Filtered out Falcon 1 launches
- Replaced null PayloadMass values with mean
- Left LandingPad nulls intact for one-hot encoding
- <https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

Insights included:

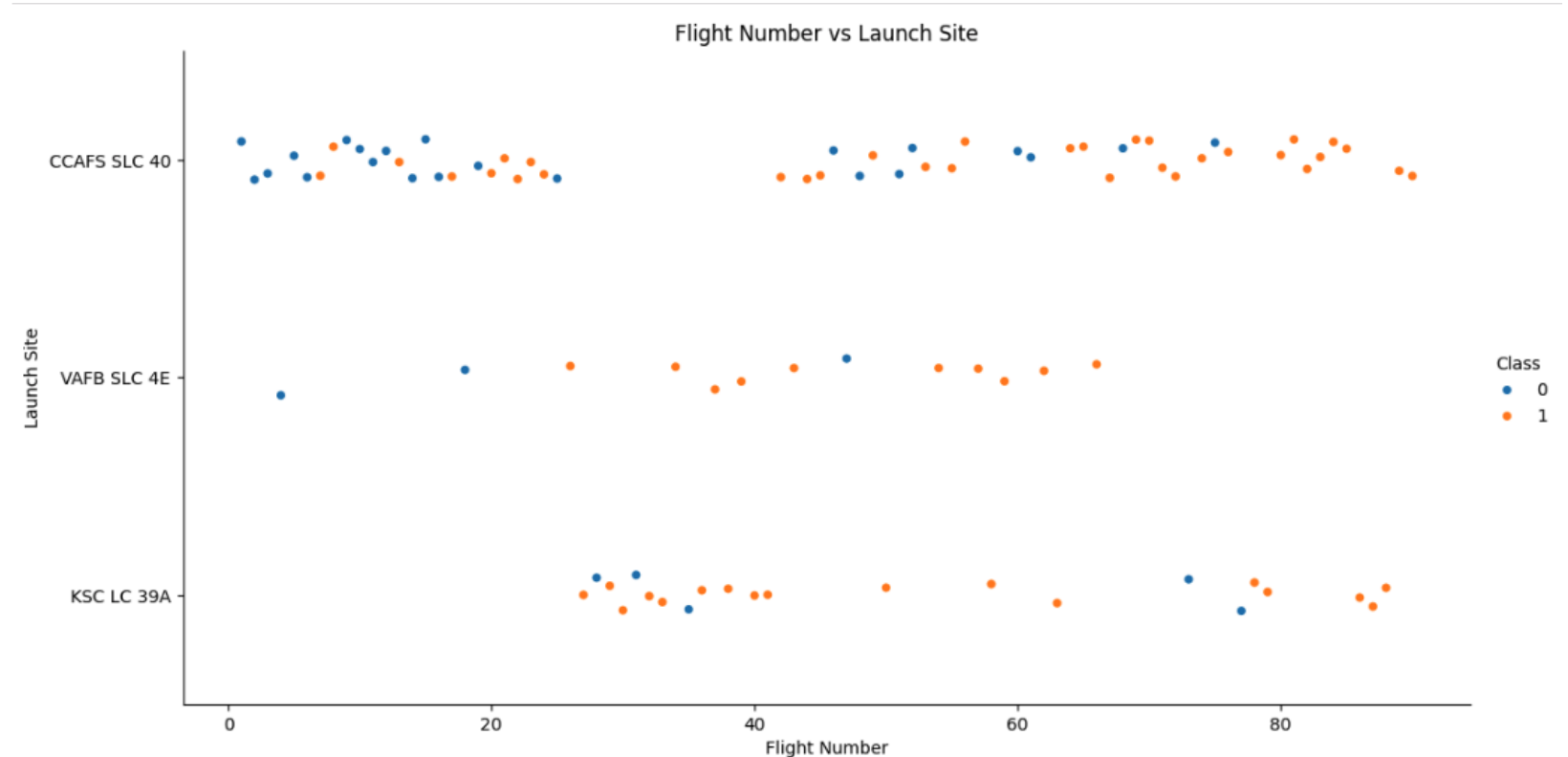
- Success rate has improved since 2013



EDA with Data Visualization

Insights included:

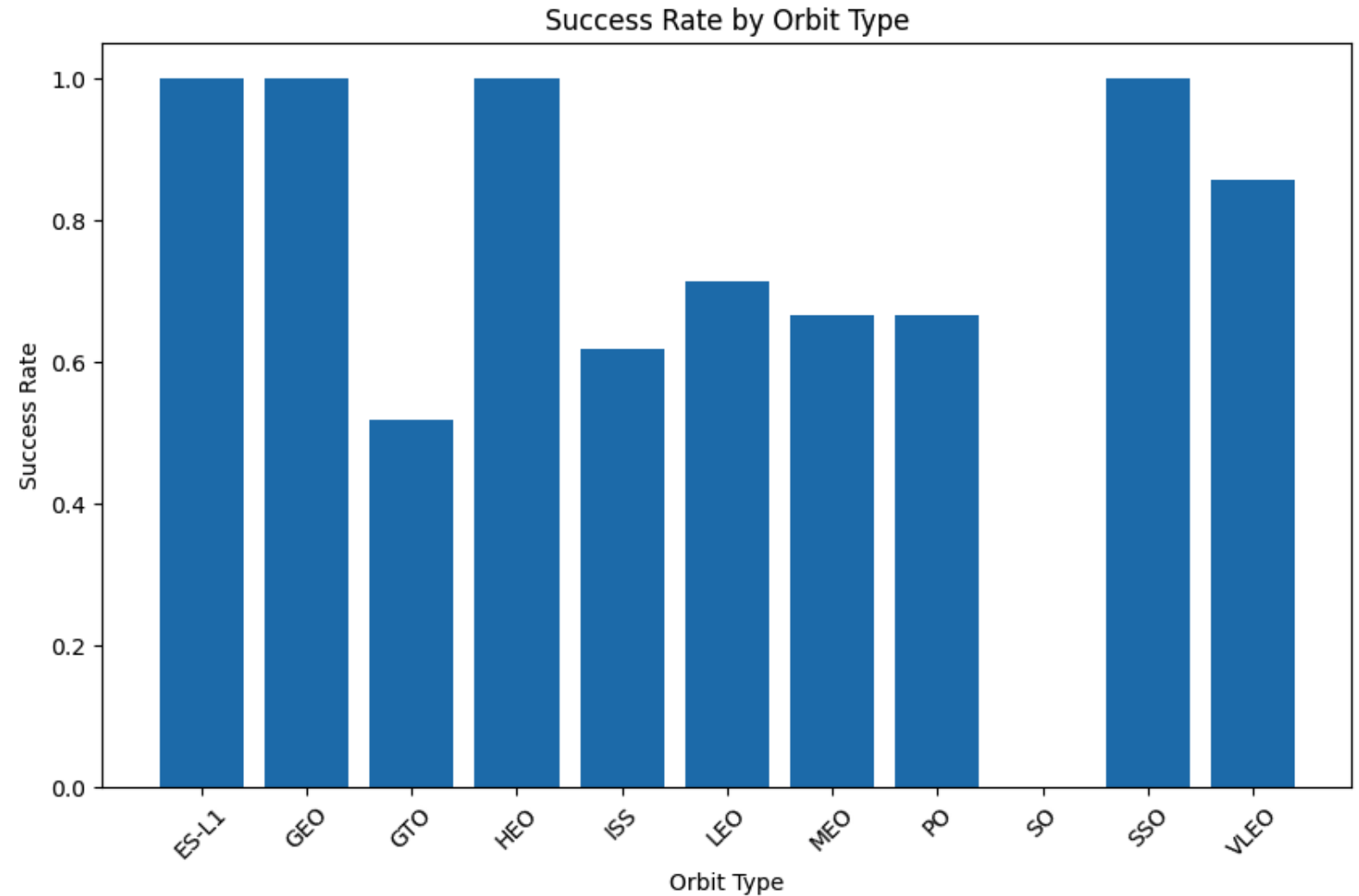
- Launch sites have different success rates
 - CCAFS LC-40: ~60%
 - KSC LC-39A & VAFB SLC-4E: ~77%
- A positive trend in mission success as experience (Flight Number) increases.



EDA with Data Visualization

Insights included:

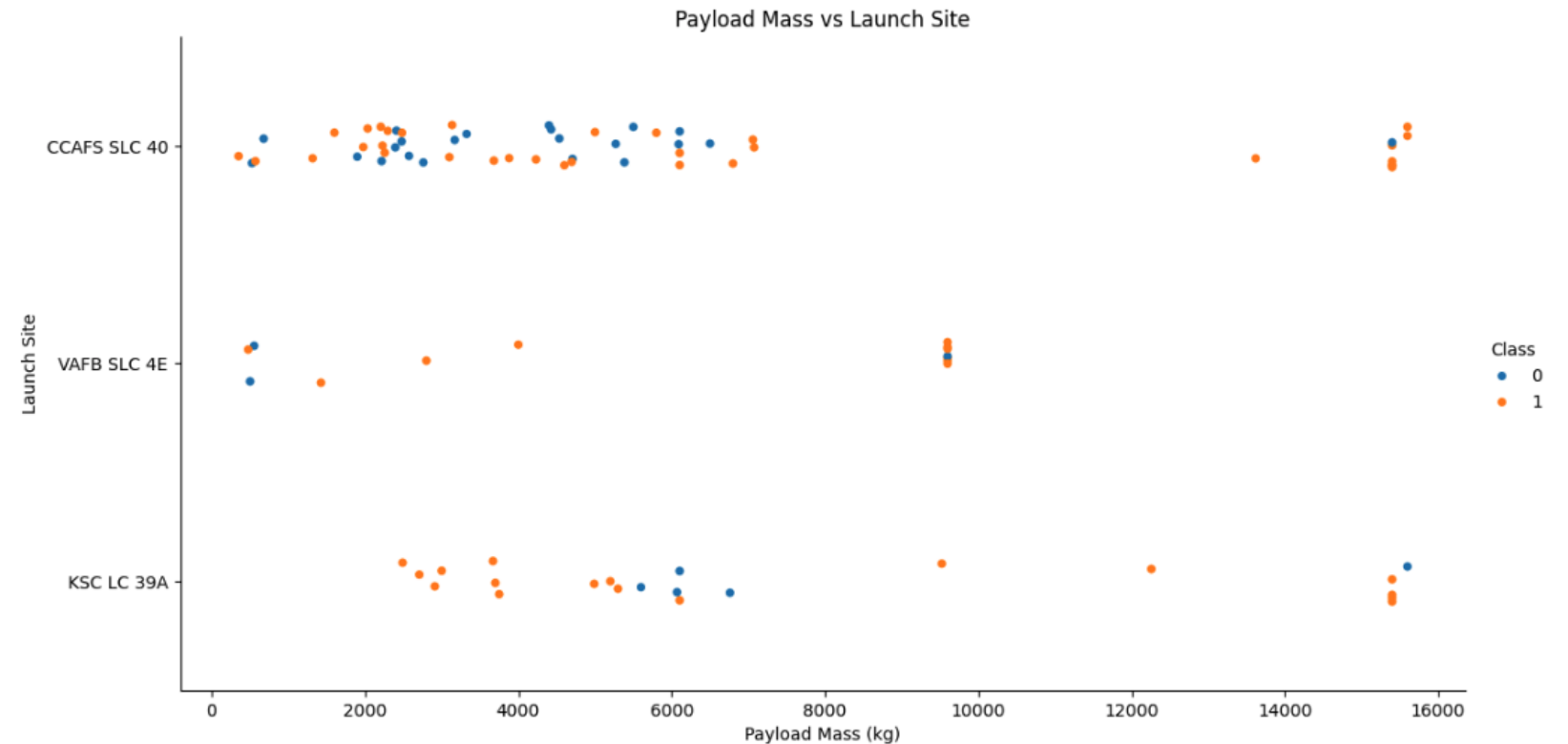
- Orbit Types have different success rates



EDA with Data Visualization

Insights included:

- Payload mass interacts with success probability
 - At CCAFS LC-40, payload > 10,000 kg → 100% success



EDA with Data Visualization

Insights included:

- Multiple features must be combined for prediction

```
features = df[['FlightNumber', 'PayloadMass', 'Orbit', 'LaunchSite', 'Flights', 'GridFins', 'Reused', 'Legs', 'LandingPad', 'Block', 'ReusedCount', 'Serial']]
features.head()
```

	FlightNumber	PayloadMass	Orbit	LaunchSite	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	6104.959412	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0003
1	2	525.000000	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0005
2	3	677.000000	ISS	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0007
3	4	500.000000	PO	VAFB SLC 4E	1	False	False	False	NaN	1.0	0	B1003
4	5	3170.000000	GTO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B1004

- <https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/main/edadataviz.ipynb>

EDA with SQL

EDA went on performing queries in SQL:

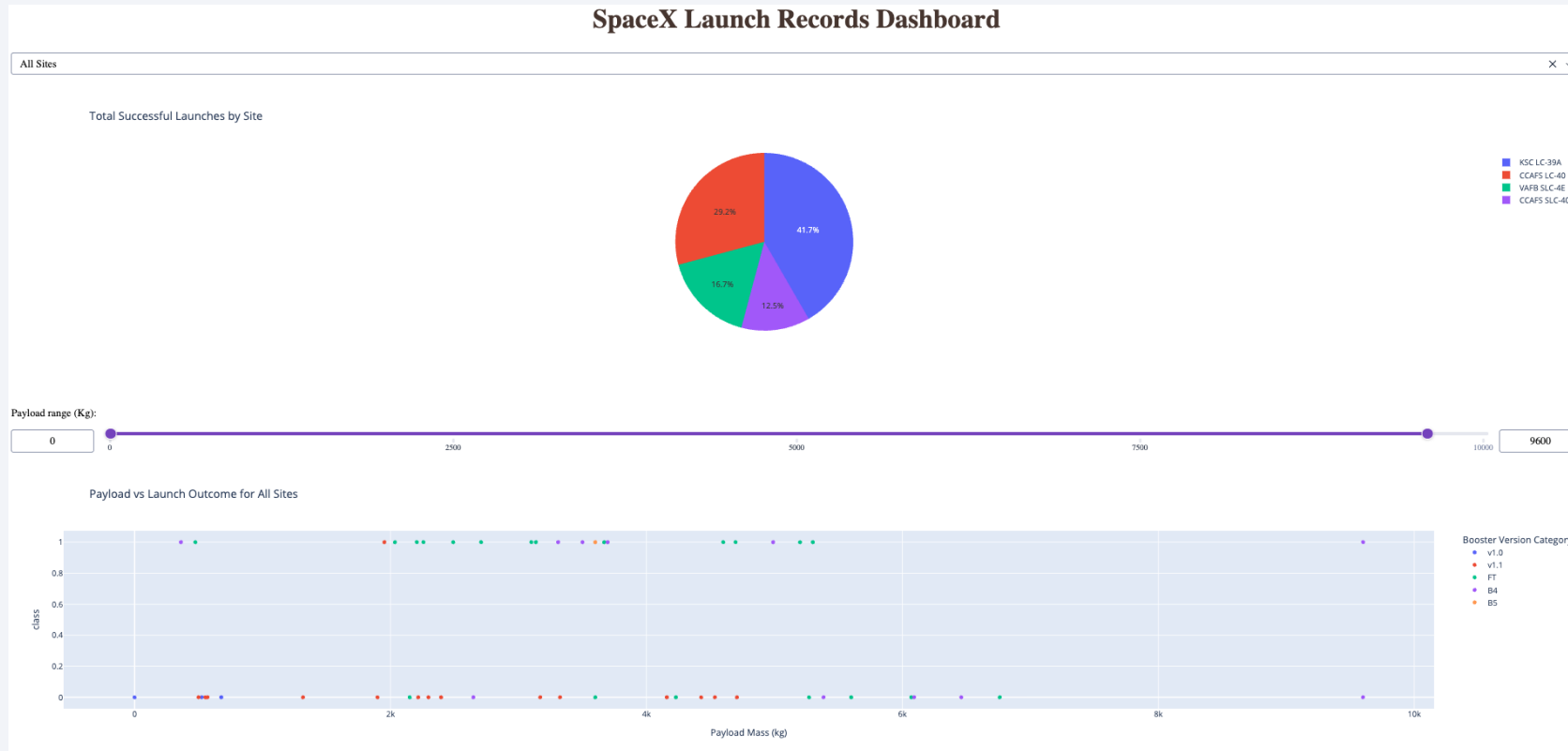
- total payload mass carried by boosters launched by NASA (CRS): 48213 Kg
- average payload mass carried by booster version F9 v1.1: 2928.4
- https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/503edadcb8a391f493052592cb64c9035a35c284/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

Object	What it is	Why it was added
<code>folium.Circle</code>	A filled circle centered on each launch site	Visually highlights each site's location on the map with a popup showing its name
<code>folium.map.Marker</code> with <code>DivIcon</code>	A text label pinned at each site's coordinate	Shows the site name directly on the map without cluttering it with default icons
<code>MarkerCluster</code>	A cluster that groups nearby markers	Prevents the map from being overloaded when many launches share the same coordinates — they collapse into a numbered cluster and expand on zoom
<code>folium.Marker</code> (green/red)	Individual launch outcome markers	Green = successful landing, red = failure — lets you visually assess which sites have higher success rates at a glance
<code>MousePosition</code>	A live coordinate display on hover	Lets you read the exact Lat/Long of any point on the map (coastline, railway, city, etc.) so you can use those values in <code>calculate_distance()</code>
<code>folium.Marker</code> with <code>DivIcon</code> (distance label)	A text marker placed at the proximity point	Displays the calculated distance (in km) right at the point of interest so the measurement is spatially anchored
<code>folium.PolyLine</code>	A straight line connecting two points	Draws a visible line from the launch site to each proximity (coastline, city, railway, highway) so the measurement is easy to interpret visually

- <https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/e8323301883530abb65daa0151225c53811a605b/lab-jupyter-launch-site-location-v2.ipynb>

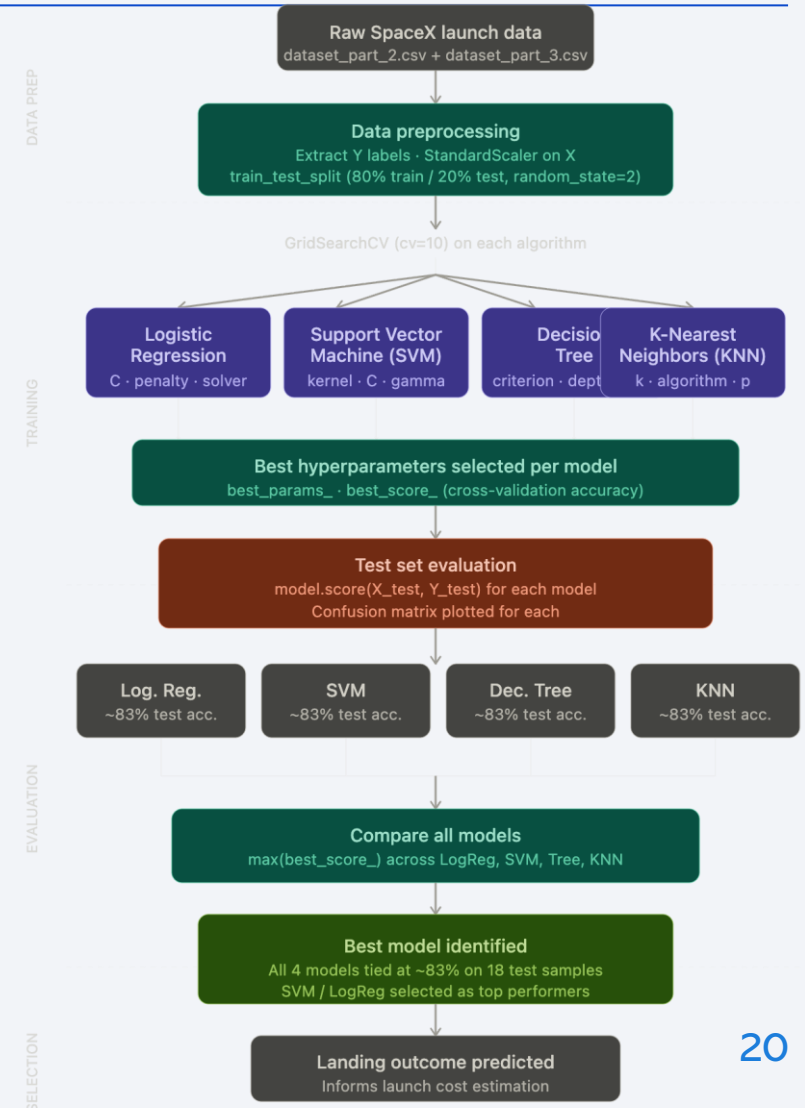
Build a Dashboard with Plotly Dash

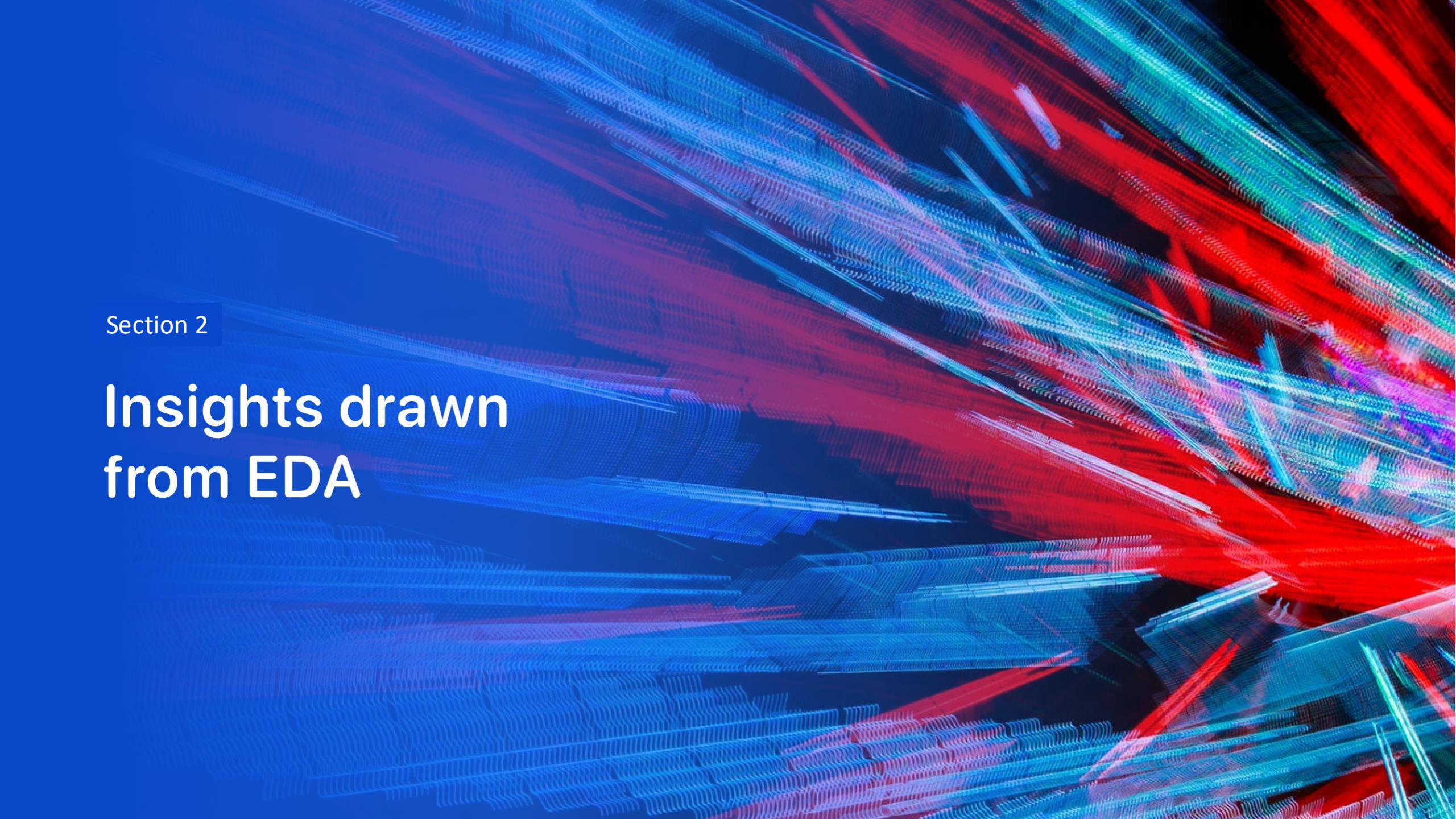


- <https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/9c08ac9eb6a37d39412f0f661d311866da8c7206/spacex-dash-app.py>

Predictive Analysis (Classification)

- **Evaluation** — Each tuned model was scored on the held-out test set using `.score(X_test, Y_test)`, and a confusion matrix was plotted for each. The confusion matrix revealed that the main failure mode across models was false positives — predicting a successful landing when it actually failed.
- **Best model selection** — All four models converged to approximately 83% test accuracy on the small 18-sample test set, making it a statistical tie. SVM and Logistic Regression are generally favored because they had the cleanest confusion matrices and are more interpretable and robust on small datasets.
- https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/blob/9c08ac9eb6a37d39412f0f661d311866da8c7206/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

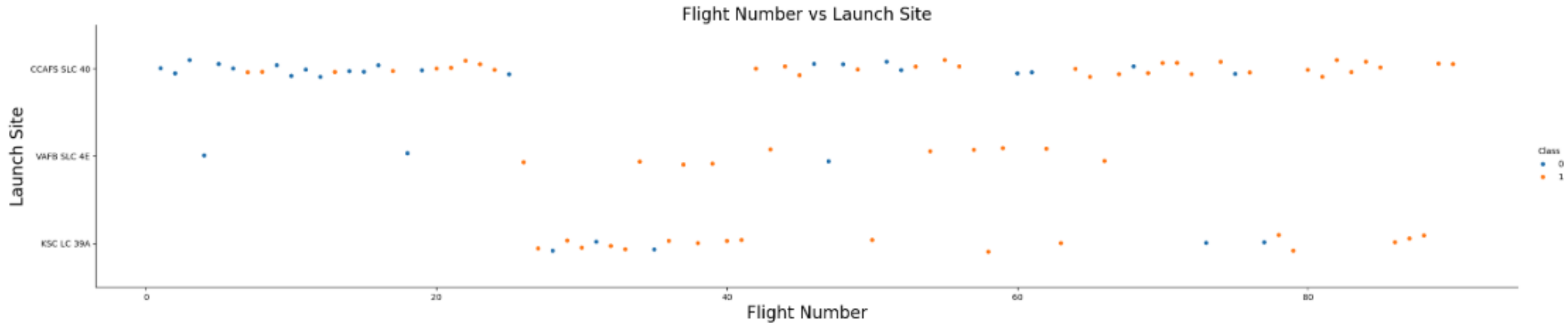


The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

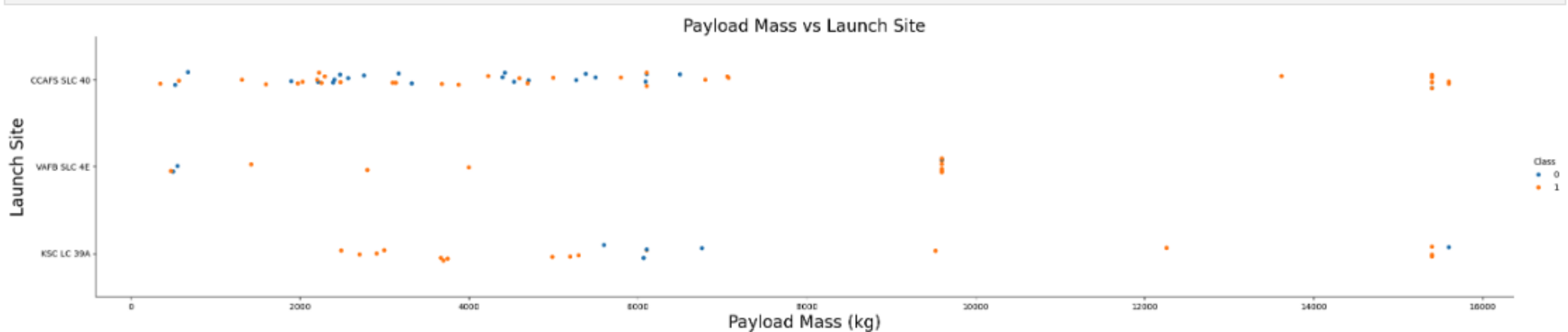
Insights drawn from EDA

Flight Number vs. Launch Site



- As flight numbers increase (i.e., later missions), the success rate (Class=1) improves across all sites.
- CCAFS SLC-40 has the most launches overall.

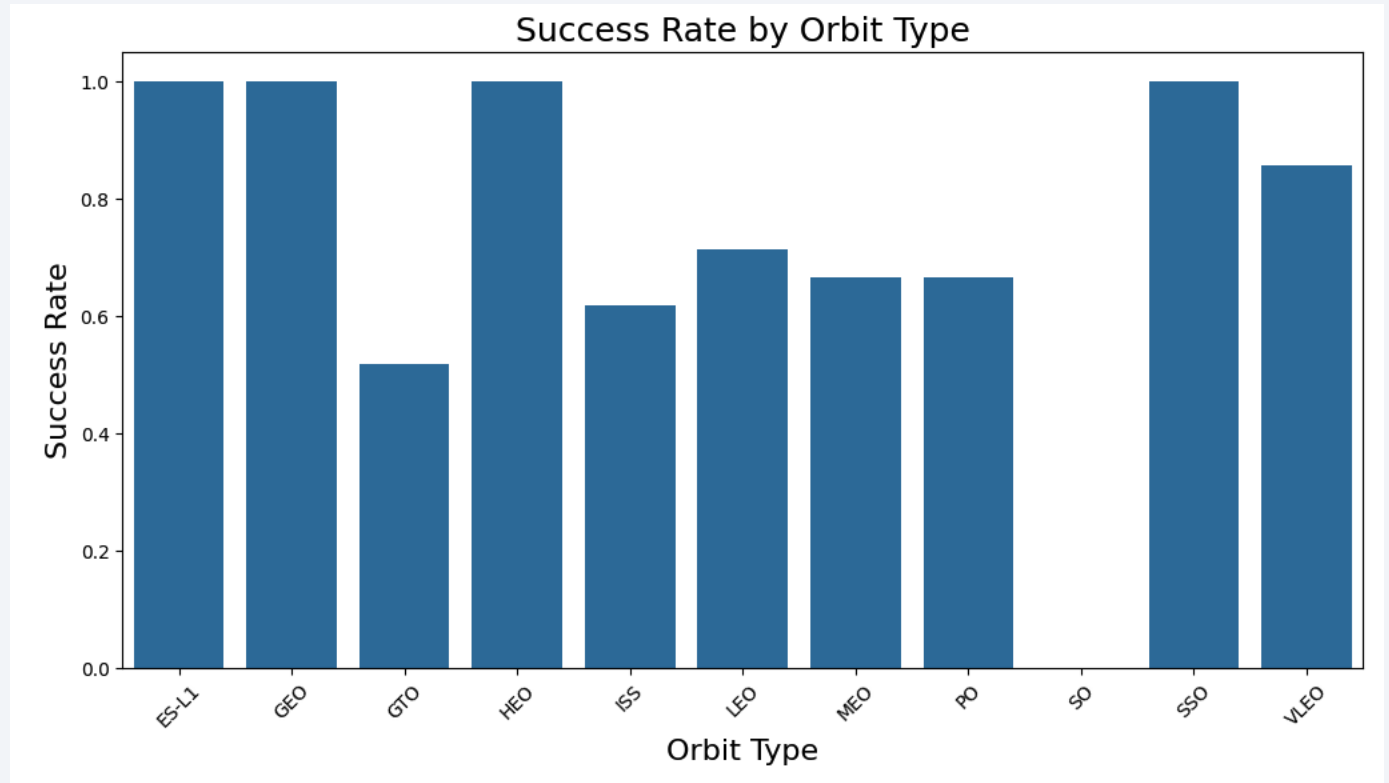
Payload vs. Launch Site



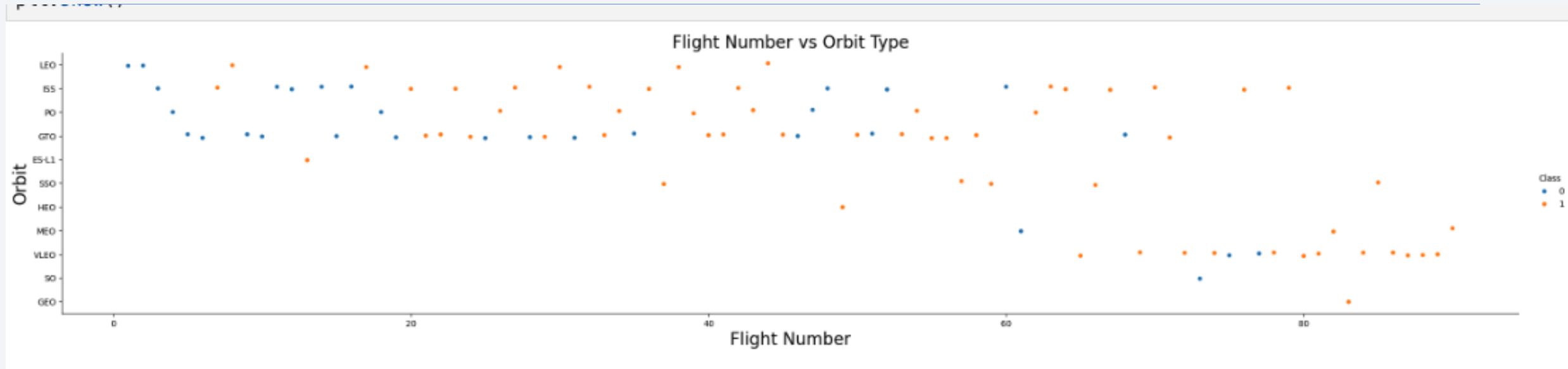
- VAFB-SLC has no launches with payload mass above ~10,000 kg.
- KSC LC-39A tends to handle heavier payloads with decent success.

Success Rate vs. Orbit Type

- ES-L1, GEO, HEO, and SSO orbits have a 100% success rate.
- SO and VLEO also perform well.
- GTO has a lower success rate due to its more demanding trajectory.

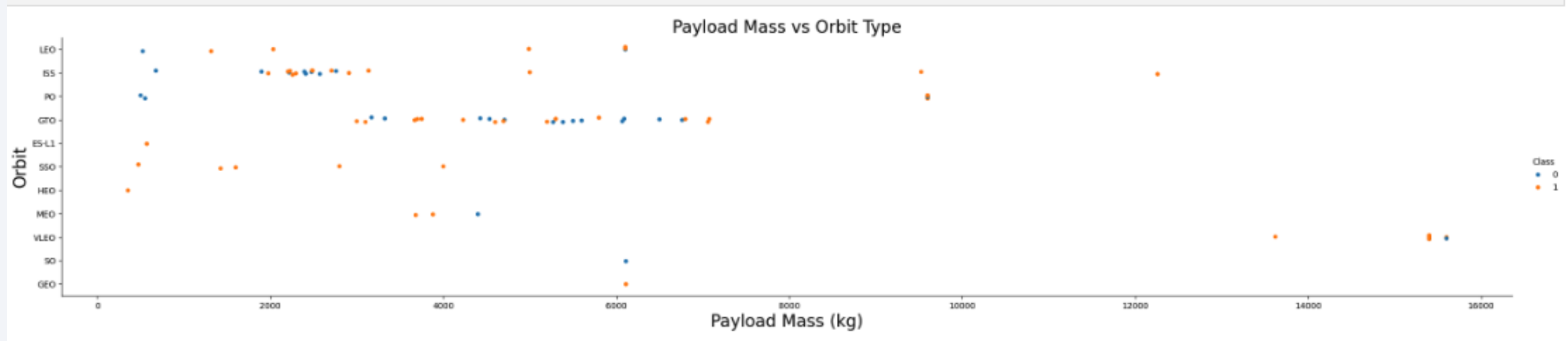


Flight Number vs. Orbit Type



- In LEO, success is strongly correlated with higher flight numbers (more experience).
- In GTO, success is scattered regardless of flight number — no clear trend.

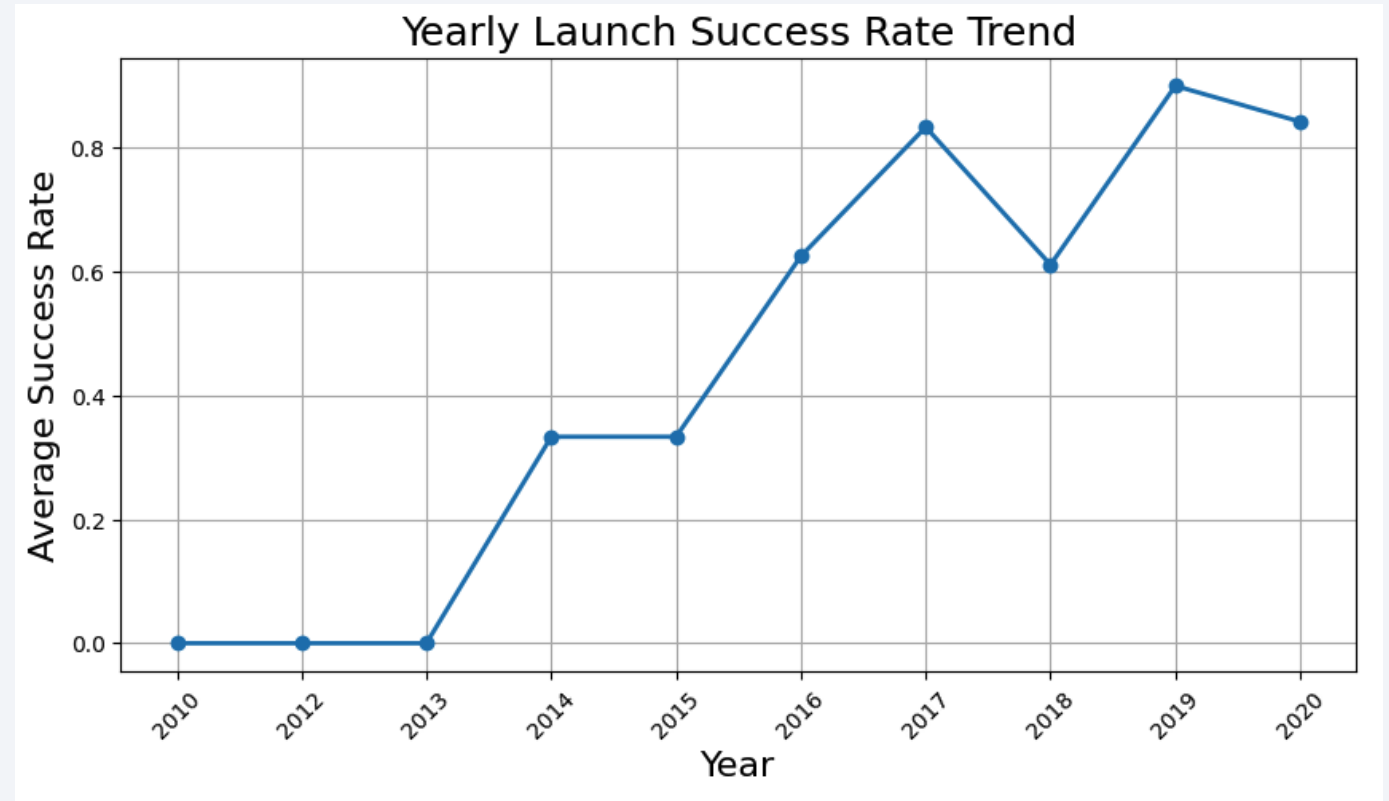
Payload vs. Orbit Type



- Heavier payloads going to LEO, ISS, and Polar orbits still tend to land successfully.
- GTO shows mixed outcomes across all payload masses.

Launch Success Yearly Trend

- Success rate was low or zero in 2010–2012, began rising from 2013, stabilized around 2014, then climbed steadily from 2015 onward, reaching near-perfect success by 2019–2020.



All Launch Site Names

- The table shows the unique launch sites names:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- The first 5 records where launch sites begin with `CCA` are:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Queries Outcomes – 1

- The total payload mass carried by boosters from NASA is 48213 Kg
- The average payload mass carried by booster version F9 v1.1 is 2928.4 Kg
- The first successful landing on ground pad was achieved on 22 Dec 2015
- Successful Drone Ship Landing with Payload between 4000 and 6000:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Queries Outcomes - 2

- Total number of successful and failure mission outcomes: 38

Landing_Outcome	total
Controlled (ocean)	5
Failure	3
Failure (drone ship)	5
Failure (parachute)	2
No attempt	21
No attempt	1
Precluded (drone ship)	1
Success	38
Success (drone ship)	14
Success (ground pad)	9
Uncontrolled (ocean)	2

- Booster which have carried the maximum payload mass

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Queries Outcomes - 3

- List of the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015:
- Ranking of the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

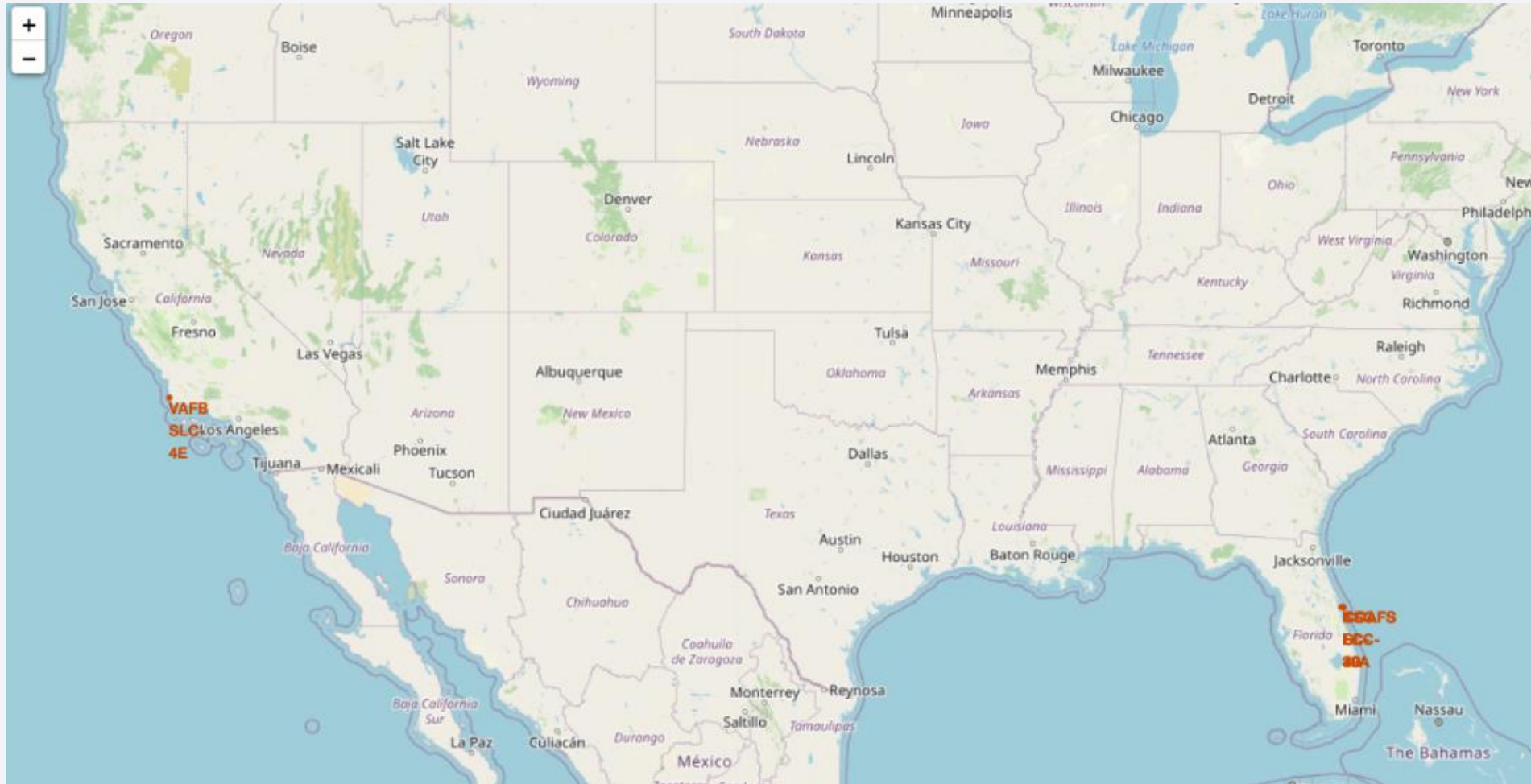
Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a thin layer of atmosphere visible along the horizon. The city lights are concentrated in the lower right quadrant, showing a dense network of urban areas. The text "Section 3" is overlaid on the left side of the image.

Section 3

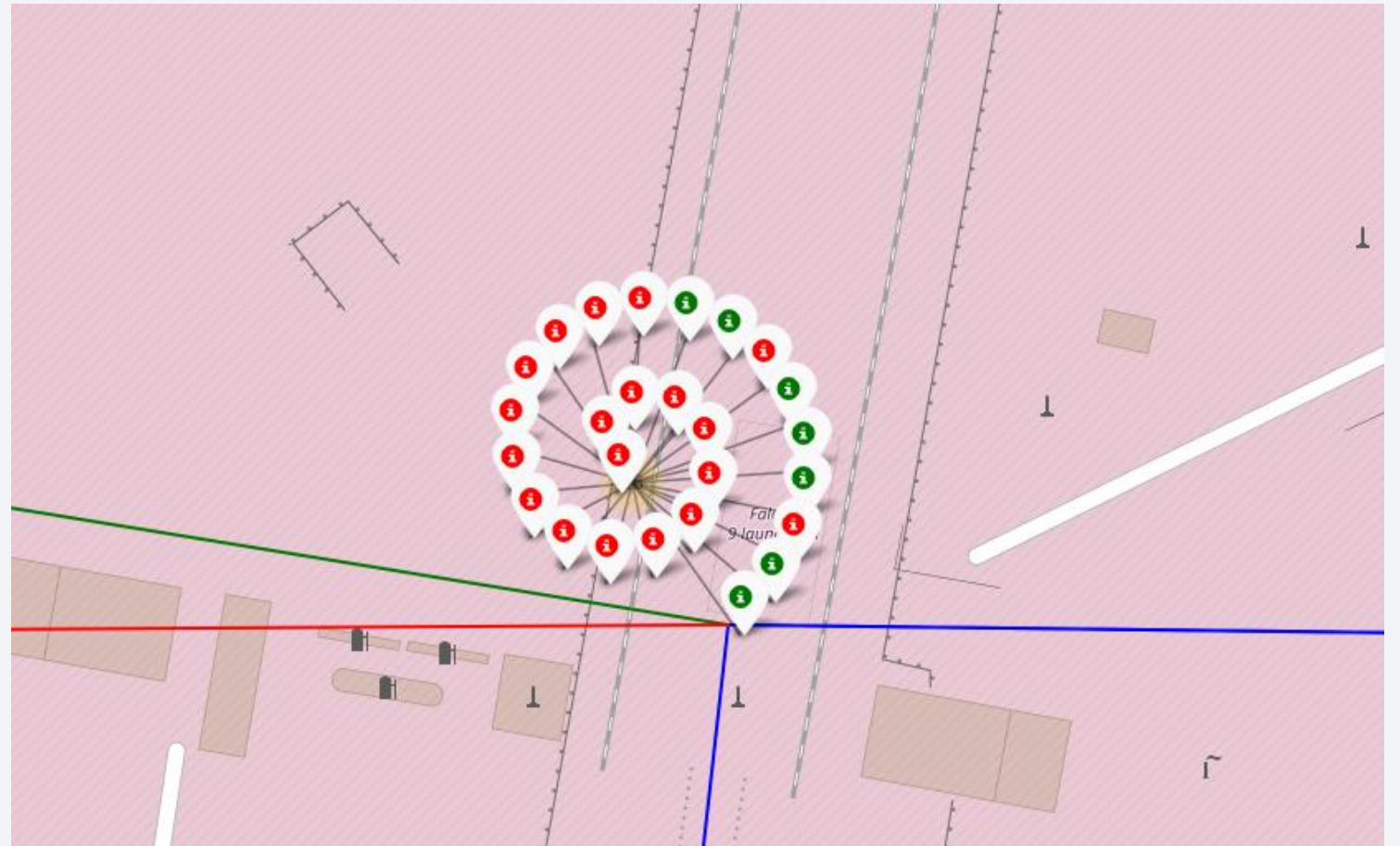
Launch Sites Proximities Analysis

Launch Sites Map



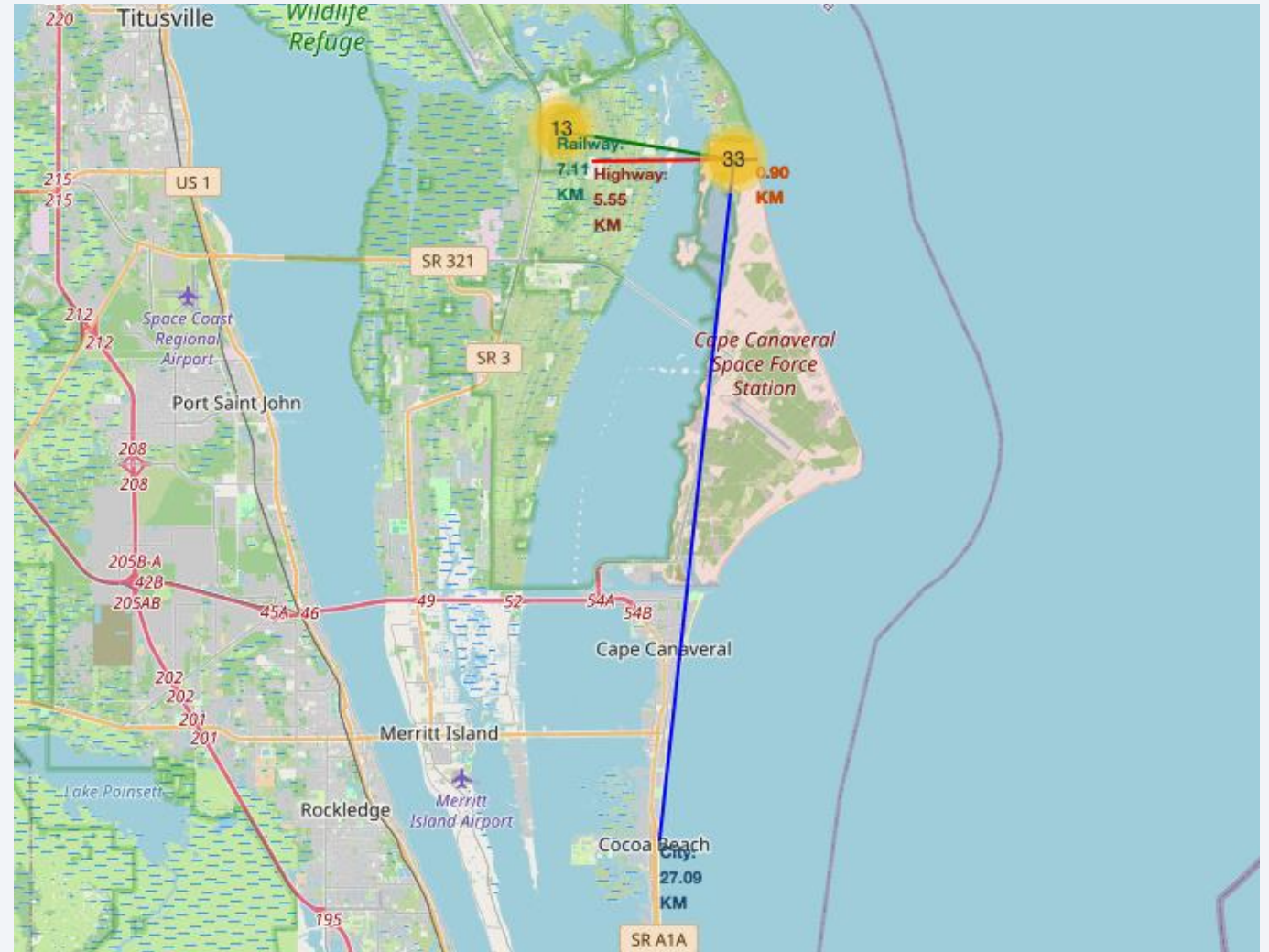
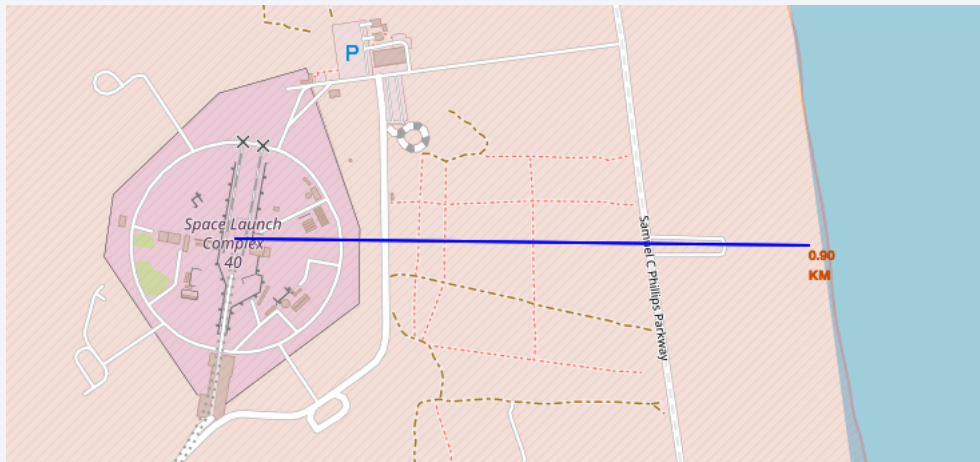
Map of success/failed launches for each site

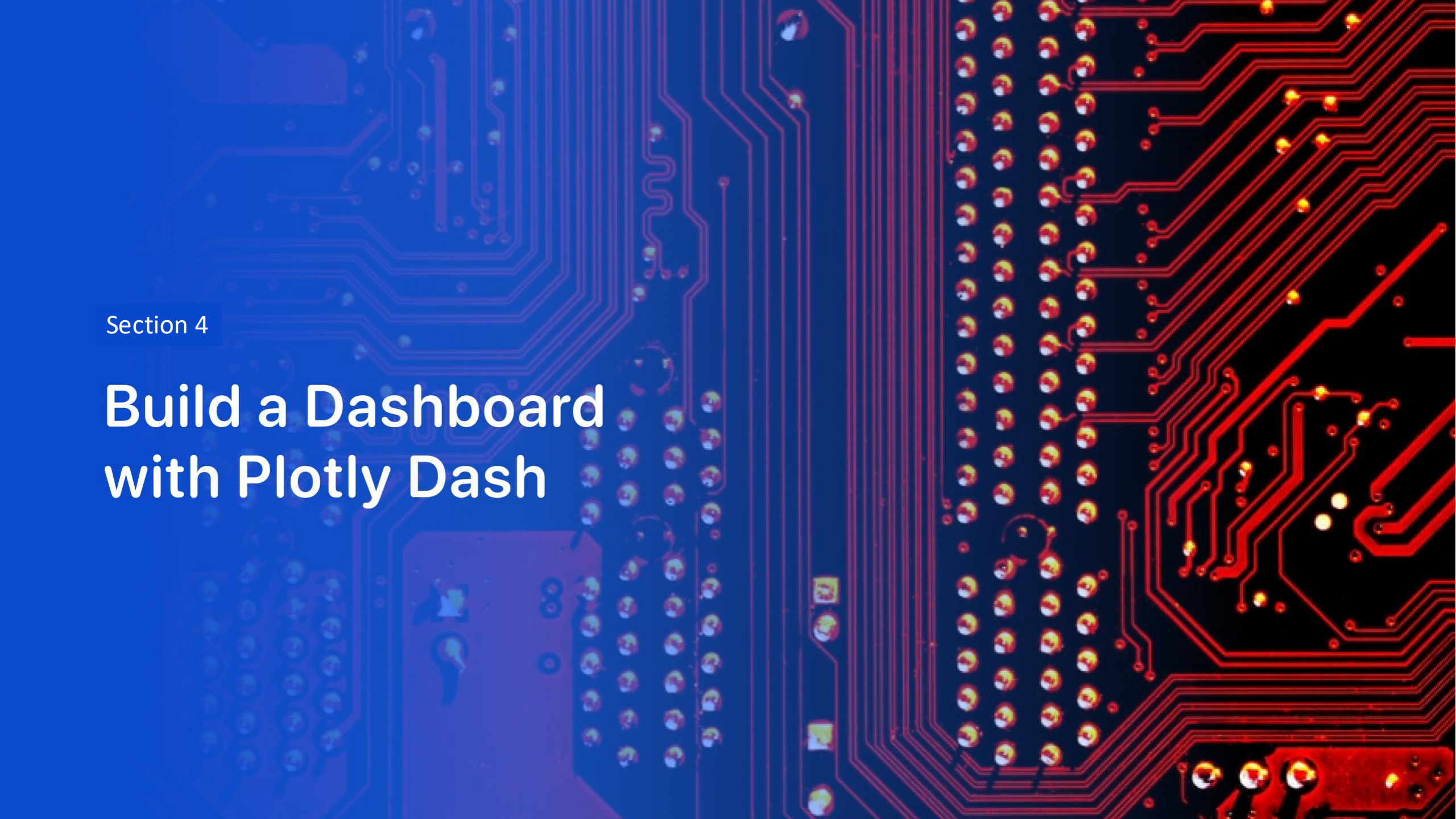
- Green: success launches
- Red: failed launches



Map of launch site proximities

- The maps highlight distances from coast and relevant sites (cities, railway, etc.).

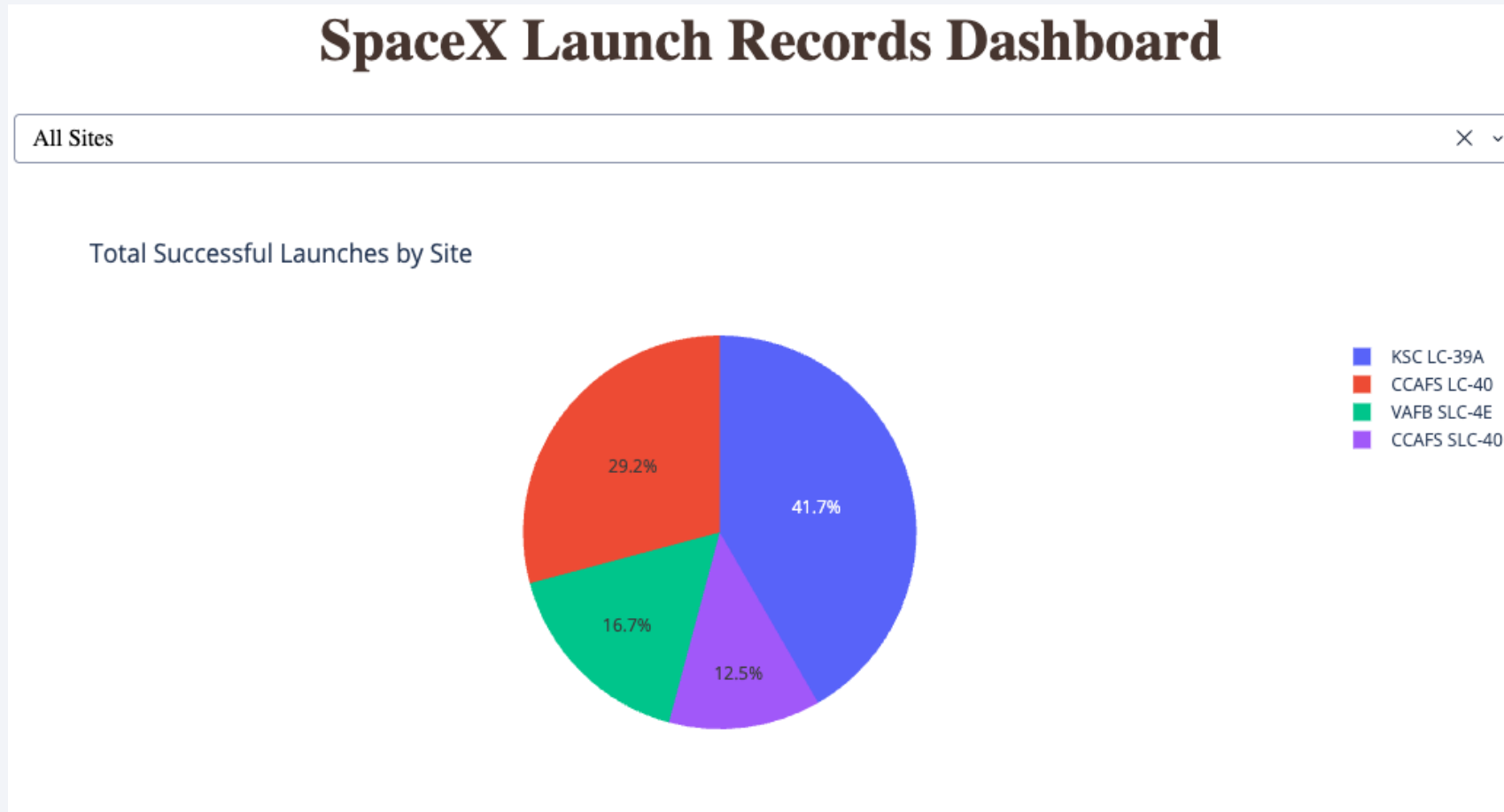




Section 4

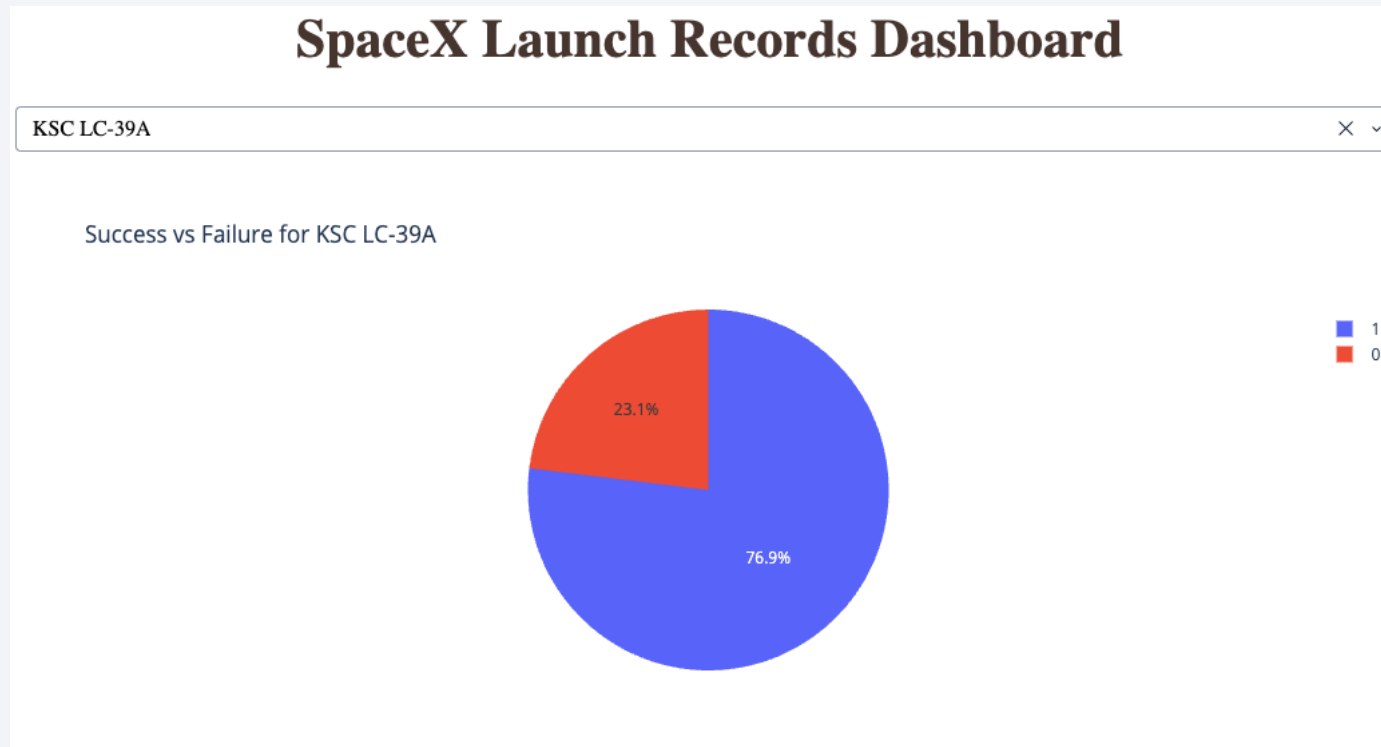
Build a Dashboard with Plotly Dash

Dashboard: Total successful launches – All Sites



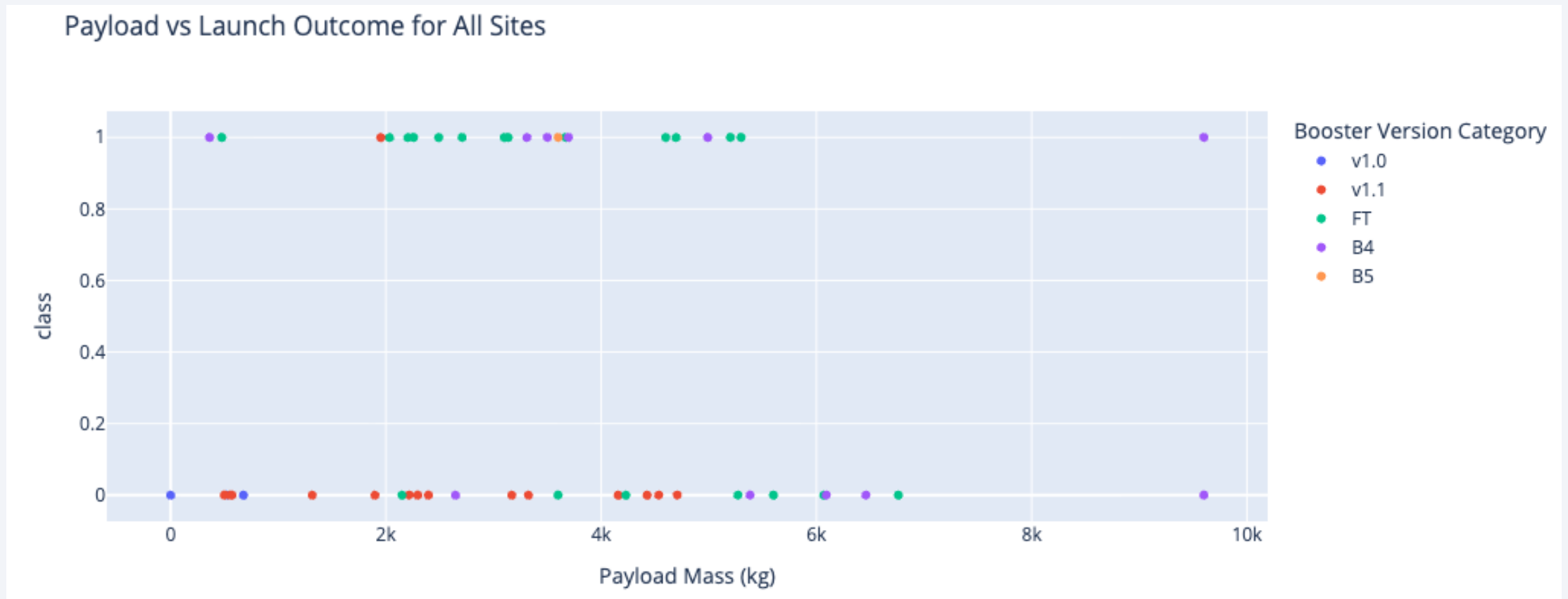
- KSC LC-39A is the site with the most successful launches

Success vs Failure for KSC LC-39A



- In the most successful launch site, success ratio is around 77%

Payload vs Launch Outcome for All Sites



- Payload Mass over 6000 Kg seems more subjected to failure
- Booster FT seems to be more successful

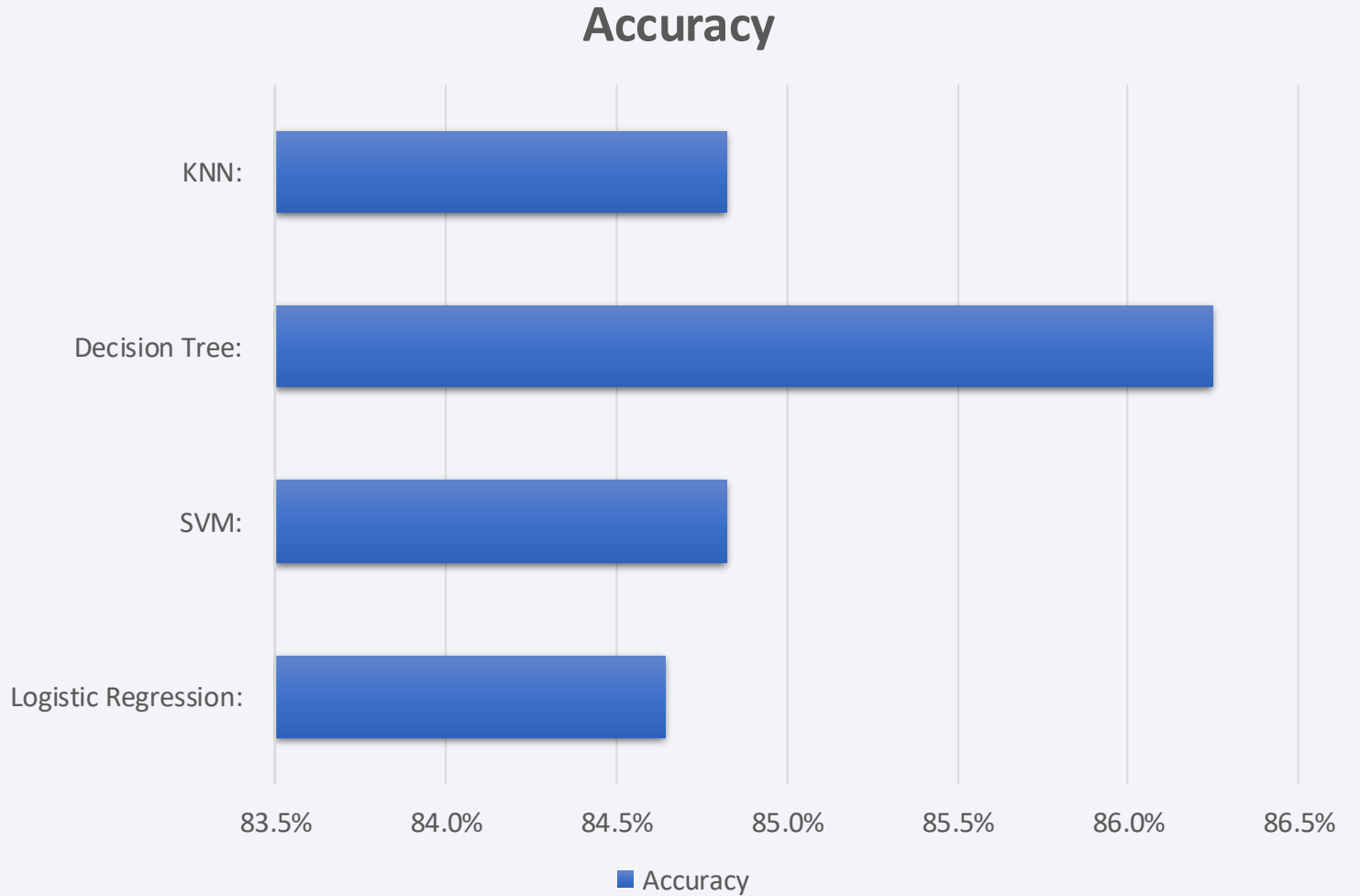


Section 5

Predictive Analysis (Classification)

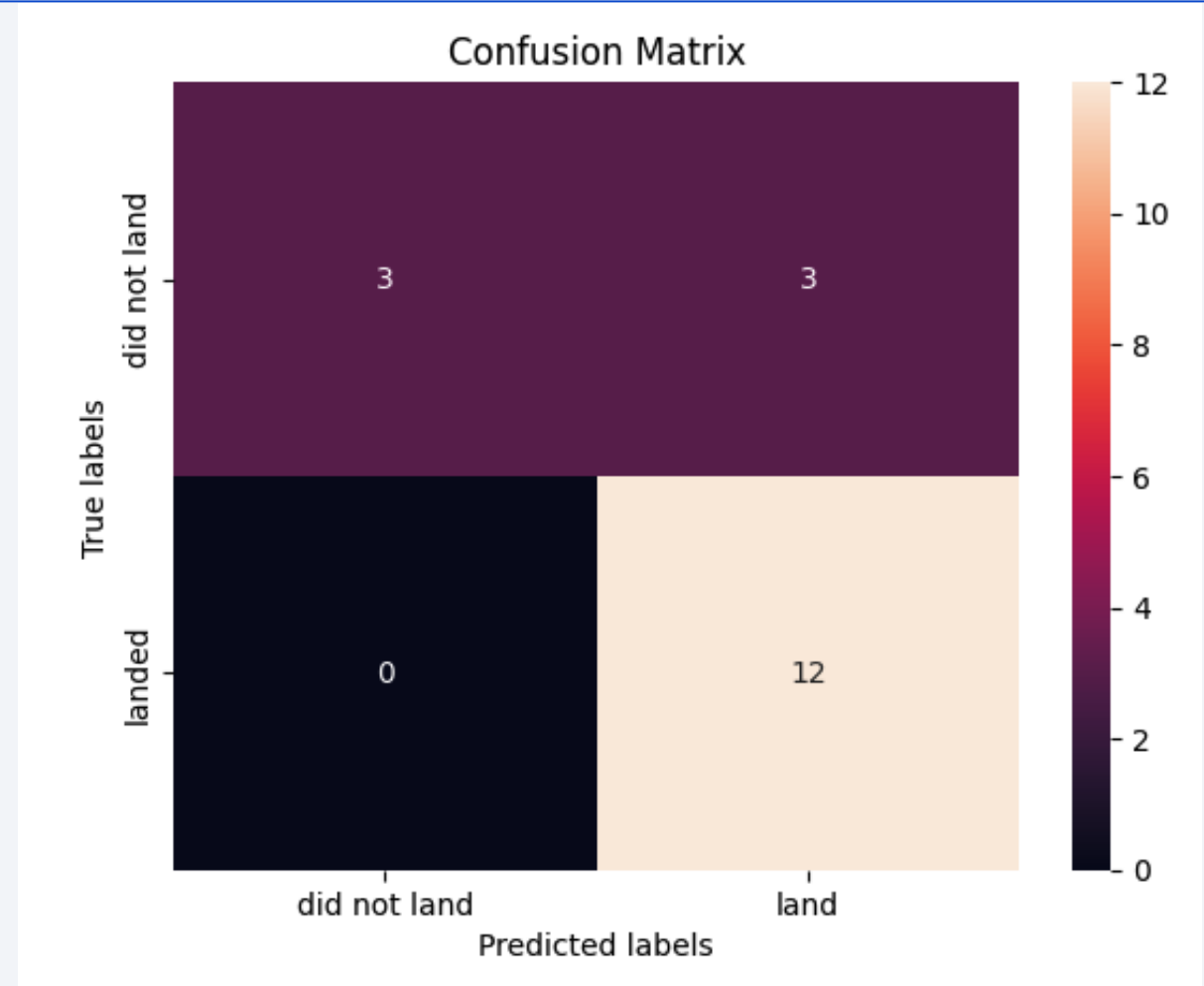
Classification Accuracy

- Decision Tree has the highest classification accuracy



Confusion Matrix

- The model is very good at detecting successful landings (zero misses), but it over-predicts success — it called 3 failed landings as successes. In a business context this matters because a false positive means incorrectly estimating a lower launch cost when the booster was actually lost.



Conclusions

- In general, all the models overestimate the success
- **The dataset is too small to draw firm conclusions:** The test set contains only **18 samples**. This is a critical limitation. The apparent tie between all four models is very likely a result of this small sample size rather than a genuine equivalence in their predictive power. A larger test set could reveal meaningful differences.
- **The class distribution is imbalanced:** Out of 18 test samples, **12 are "landed" (class=1)** and only **6 are "did not land" (class=0)**. This 2:1 imbalance favors models that default toward predicting success, which partly explains why all models achieve 83% accuracy and why false positives are the main failure mode rather than false negatives.
- **The problem is learnable — the signal is real:** Despite the small dataset, all models consistently outperform a naive. Reaching 83% means the features — payload mass, orbit type, launch site, booster version, flight number — do carry genuine predictive signal about landing outcomes.
- **Business implication:** A model that can reliably predict landing success allows **competing companies to make informed bids**. Even at 83%, the model is a useful decision-support tool, though the false positive rate means it should not be used as the sole basis for cost estimates.

Appendix

- Link to Github Repo:
 - <https://github.com/RaffaeleBini/Applied-Data-Science-Capstone/tree/main>

Thank you!

